

DECATUR, IL, USA

WMO: 725316

Lat: 39.834N

Lon: 88.866W

Elev: 675

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	6.7	-6.2	4.0	3.7	-1.8	5.1	7.4	27.8	29.9	25.6	32.2	9.6	300	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	92.6	76.3	90.4	75.3	88.1	74.2	79.2	89.0	77.7	87.0	76.2	85.0	10.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	140.6	85.7	74.8	133.6	83.9	73.2	126.4	81.9	43.2	89.2	41.5	86.9	40.0	84.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.4	19.3	-5.3	96.5	7.0	2.8	-10.4	98.5	-14.5	100.1	-18.5	101.6	-23.6	103.6		
			-6.0	82.2	6.8	1.5	-10.9	83.2	-14.9	84.1	-18.7	84.9	-23.7	86.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	27.4	31.6	42.0	53.4	63.9	72.6	75.7	73.9	67.6	55.6	42.9	32.3
	DBStd	19.28	12.31	11.53	11.06	9.24	8.42	6.31	5.64	5.39	7.44	9.11	9.98	10.87
	HDD50	2460	705	523	301	66	4	0	0	0	1	45	259	556
	HDD65	5374	1166	935	717	361	127	13	1	3	56	317	665	1013
	CDD50	3683	4	7	51	169	436	678	798	742	529	217	45	7
	CDD65	1124	0	0	3	13	93	241	334	280	134	25	1	0
	CDH74	10971	0	0	21	142	834	2294	3429	2625	1378	244	4	0
	CDH80	3991	0	0	1	19	227	839	1377	969	501	58	0	0
Wind	WSAvg	9.6	11.5	11.4	11.5	11.8	9.9	8.2	7.0	6.6	7.1	9.3	10.7	10.6
Precipitation	PrecAvg	39.5	2.2	2.0	2.6	3.9	4.6	4.6	4.1	3.4	3.1	3.2	3.5	2.1
	PrecMax	51.3	6.1	4.8	5.5	8.5	10.2	9.3	12.7	5.4	8.2	7.1	8.2	4.3
	PrecMin	33.9	0.4	0.5	0.8	1.3	1.3	2.3	1.5	1.5	0.2	0.3	0.9	0.9
	PrecStd	4.9	1.3	1.1	1.2	1.9	2.6	2.0	2.6	1.0	2.1	1.7	1.7	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.2	65.5	77.2	82.3	89.4	93.2	97.2	95.1	93.0	86.4	72.8	63.8
		MCWB	57.7	56.0	63.9	65.6	72.2	75.1	78.7	77.2	72.3	70.0	60.8	59.6
	2%	DB	55.3	59.8	70.9	78.6	85.5	90.7	92.9	91.4	89.7	81.0	67.0	58.1
		MCWB	52.0	54.3	60.0	63.3	70.0	74.6	77.7	76.0	72.1	66.2	58.8	54.5
	5%	DB	50.4	54.1	65.3	73.9	82.3	88.3	90.3	88.7	85.8	75.8	63.3	53.1
		MCWB	46.8	49.0	56.7	61.0	68.4	74.0	77.1	75.0	70.3	63.5	56.9	49.4
	10%	DB	44.0	48.1	60.1	70.1	79.1	85.2	87.6	85.6	82.0	71.7	58.9	47.5
		MCWB	40.4	43.6	52.8	59.2	66.4	71.9	75.1	73.4	68.2	62.0	52.7	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	59.6	65.3	69.2	75.1	79.2	81.8	81.0	76.6	72.2	64.2	60.5
		MCDB	61.5	63.5	75.0	79.7	85.0	89.1	92.3	91.0	87.0	83.1	69.1	63.6
	2%	WB	53.1	54.8	61.8	66.0	72.5	77.3	79.9	78.4	74.6	69.1	61.0	55.1
		MCDB	55.0	58.6	69.4	73.7	82.2	87.6	89.8	88.1	84.8	77.1	65.7	57.9
	5%	WB	46.5	49.1	58.2	63.4	70.5	75.6	78.3	76.9	72.5	66.3	58.0	49.4
		MCDB	49.4	54.1	64.1	70.7	79.4	85.1	87.5	85.8	81.9	72.9	62.5	52.3
	10%	WB	40.4	43.8	53.5	60.6	68.4	74.0	76.9	75.1	70.5	63.1	53.5	44.4
		MCDB	43.9	48.1	59.1	68.4	76.2	82.6	85.4	83.1	78.9	70.3	58.2	47.3
Mean Daily Temperature Range	5% DB	MDBR	14.8	16.1	18.8	21.1	20.1	19.5	19.6	20.3	23.3	20.8	17.1	13.9
		MCDBR	20.3	23.6	25.2	26.1	22.9	22.1	21.1	22.9	26.2	26.3	22.7	19.5
		MCWBR	17.4	18.4	16.5	15.1	11.3	10.2	9.6	10.5	11.7	14.3	16.4	16.4
	5% WB	MCDBR	19.3	22.5	22.4	22.2	20.3	19.8	19.1	19.5	21.7	21.5	20.4	18.1
		MCWBR	17.7	18.8	15.8	15.1	11.4	10.2	9.9	10.2	11.0	13.6	16.7	16.6
Clear-Sky Solar Irradiance	taub	0.303	0.315	0.352	0.392	0.436	0.435	0.460	0.453	0.411	0.360	0.312	0.297	
	taud	2.440	2.397	2.375	2.265	2.195	2.239	2.154	2.164	2.261	2.408	2.497	2.472	
	Ebn at Noon	274	287	286	280	269	268	260	259	262	266	269	268	
	Edh at Noon	25	30	35	42	46	44	47	46	39	30	24	23	
All-Sky Solar Radiation	RadAvg	580	852	1161	1523	1734	1992	1967	1748	1477	1022	682	488	
	RadStd	67	83	92	102	146	137	117	112	106	111	66	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+126

Nomenclature: See separate page